

Practical application of RINO, smartphone-based dynamic displacement sensing application, for wind tunnel tests

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ABSTRACT

Dynamic displacement is one of the most important measurands in wind tunnel tests of structures. Laser sensors or optical sensors are usually used in wind tunnel tests to measure displacements. However, these commercial sensors have limitations in its use, cost and installation despite of their good performance in accuracy. RINO (Real-time Image-processing for Non-contact monitoring), an iOS software application for dynamic displacement monitoring, has been developed in the previous study. In this study, feasibility of RINO in practical use for wind tunnel tests is explored. Series of wind tunnel tests show that performances of RINO are comparable with those of conventional displacement sensors.

Keywords: Displacement sensor, aerodynamic, wind-tunnel test, smartphone sensor

1. INTRODUCTION

Measurement of static and dynamic displacement is particular important both for field tests and for laboratory tests. However, current Structural Health Monitoring (SHM) technologies rarely support displacement monitoring, primarily due to the difficulty in measuring absolute displacement. Wind tunnel tests also benefit from displacement measurements. Laser displacement sensors or optical position sensors are generally used in wind tunnel tests [1-3]. However, the high costs and restricted use of these conventional sensors to cope with various measuring situations poses technical and practical challenges in wide-spread applications.

Computer vision-based methods have received broad interests in measuring static and dynamic displacements of structures due to various merits of the methods. These vision-based measuring systems are available at relatively low cost, combining a video camera, an optical zoom lens, lighting lamps and a precision target attached to the location of interest on the structures [4].

In the previous study, an iOS software application, named RINO (Real-time Image-processing for Non-contact monitoring) has been developed both for field and laboratory applications to measure dynamic movements of targets. The newly developed smartphone application allows measuring absolute dynamic displacements in real time using state-of-art smartphone technologies, such as high-performance graphics processing unit (GPU), in addition to already powerful CPU and memories, embedded high-speed/resolution camera, and open-source computer vision libraries [5].

In this study, use of RINO for practical applications is explored, with particular emphasis on wind tunnel application. RINO has been used for wind tunnel tests to check the stabilities of a bridge deck against torsional flutter and vortex-induced vibration. Various sensing setups available for RINO have been tested and optimized under same condition with conventional measurement setup. And the performances of RINO have been compared with those from conventional displacement sensors. Two-types of measurement were tested to figure out the performance of RINO. One was same condition with conventional sensor measuring two target using two sensors. The other was measuring a single target by a single RINO. Both of results show good accuracies comparing with conventional measurement.

2. RINO: SMARTPHONE APPLICATION FOR REAL-TIME IMAGE PROCESSING FOR NON-CONTACT DYNAMIC DISPLACEMENT MONITORING

RINO is an iOS app developed to measure dynamic displacement using smartphone vision technologies. There are several difficulties to measure absolute displacement of structures by conventional methods. Because most of existing technologies for structure displacement measurement, no matter what types are used (i.e. contact-type, non-contact type but short range) require stationary reference points nearby the target structure. And relatively low sampling rates and low-accuracy [6-10] limit the use for wide-spread field applications of the sensors. Despite there are high-speed video cameras that can measure up to 2000 fps with acceptable accuracy [11]. However, these types of cameras are too expensive to use for civil engineering applications.

Harnessing smartphone technology provides many advantages for SHM. Recent advances in smartphone technologies allow high-resolution and high-speed frame rate camera, powerful processors and substantial memories of the smartphones enable onboard image processing and large data storage. Moreover, as long as smartphones are available nearby, no much additional cost and equipment are necessary, making it ideal for short-term, campaign-type, portable dynamic sensing for structures, as opposed to conventional, expensive, and complicated dynamic sensors.

RINO has been developed for iPhone on Open GL (Open Graphic Library) in the iOS environment. Comparing with another branded smartphone such as Galaxy S6 and LG G3 which have lower camera frame rate (30~60 fps) than iPhone [12-13], iPhone 6 has higher frame rate (up to 240 fps) [14]. Having such high-frame rate (i.e. sampling rate) is critical in preventing aliasing issue in dynamic sensing. Availability of such high-frame rate of iPhone was one of important reasons to develop RINO in the iOS environment for iPhone first. RINO for android will be further developed.

RINO has several functionalities. i) Real-time processing - dynamic displacement is calculated and displayed in real time without saving too much image for post processing. ii) Crop filter - user can define measurement area to reduce processing time. Figure 5 shows RINO app running on the iPhone 5s to monitor dynamic motion of a target with various types of crop filter, 200×400 , 400×400 and 200×720 crop filter. It confines measurement area which improves calculation performance drastically. iii) Target detection filter - this filter detects optimized blue-yellow/purple-green color pattern by default and user define color pattern. In this filtering process, hue range is used rather than RGB range because with small change of color value, hue values of color slightly change in contrast with can dramatically change RGB values. iv) Adjustable frame rate - user can choose video sensing frame rate among 30, 60, 120 and 240 fps depends on the characteristics of target movements. v) Latest version of iPhone 6s supports up to 240fps at 720p. vi) Additionally, RINO offers a high-performance mode that turn off the screen while testing to secure better calculation performance rate. vii) Onboard calibration - RINO calculates the movements of target in engineering displacement unit using *GPUImage3x3ConvolutionFilter* which identify colors from target and its centroid is calculated by CPU at any given frame. viii) Data exporting - user can email measured data as CSV that allows convenient data acquirement.

Displacement measurement process by RINO is as follows. Firstly, Camera captures the target and GPU displays to screen. Then, crop filter works to define measuring area and identifies two-targets and calculate centroid position in the measuring boundary. Finally, update the graph. Dynamic displacement on the app is updated under repetition processes at any given frame.

3. WIND TUNNEL TEST FOR BRIDGE DECK

3.1 Section Model Test for Bridge Deck

Stability for wind load is a very important factor, particularly, in long-span bridge design. Therefore, several wind tunnel tests are conducted to find optimized sectional shapes in most of long-span bridge design process.

In case of Akashi Kaikyo Bridge, for example, a truss type girder section has been selected for main girder of the bridge according to the results from series of wind tunnel tests with various types of sections, such as single box and twin box [15]. For the case of Great belt bridge, a single box-based wind-resistance test for edge angle has been conducted [16]. Similarly, most long-span bridge sections were selected based on wind-tunnel tests. Thus, sectional model test is necessary to secure the wind-resistance of long-span bridges. Compared with full bridge model tests, sectional model tests have several advantages; which include i) lower cost in model manufacture and tests, ii) easy to replicate with detail and revisable. Furthermore, a sectional model test is simpler, takes shorter time for test, so that is appropriate for checking the stabilities of various type of sections and finding optimized sectional shapes.

3.2 Wind tunnel facility

The section model tests were conducted in the wind tunnel of TESolution Co., Ltd, in Anseong-si, South Korea. As one of the world's top leading companies in the fields of wind engineering and vibration control, TESolution has good facilities for various wind tunnel and vibration control experiments and has more than 150 wind tunnel study experiences for long-span bridges in last twelve years. The main specifications of wind tunnel used for the sectional model test are as follows:

- Type: Eiffel type
- Size of test section: 1m(W) x 1.5m(H) x 6.0m(L)
- Spring Support System: Controls of vertical/torsional natural frequency, attack angle, mass, mass moment of inertia are possible.
- Wind Speed: 0.3 - 21.0 m/s
- Turbulence Intensity: Less than 0.7%
- Flow Uniformity: Less than 1.0%



Figure 1. Eiffel-type wind tunnel for section model tests

A spring support system, installed at the outside of the wind tunnel so as not to block the wind flow (see Figure. 2), is used in the section model test to measure aerodynamic response. The section model is mounted on the spring support system so that its longitudinal direction is orthogonal with respect to the flow and model's vertical oscillation. The supporting system permits the section model to oscillate in the respective torsional and vertical direction as well as in the coupled manner of two oscillations. Reproducing dynamic properties such as mass, stiffness and damping of the section model to the target values can be achieved by properly adjusting the components of the spring system and damping device. In addition, upwind incidence angle can be manipulated with mechanical adjusting system.

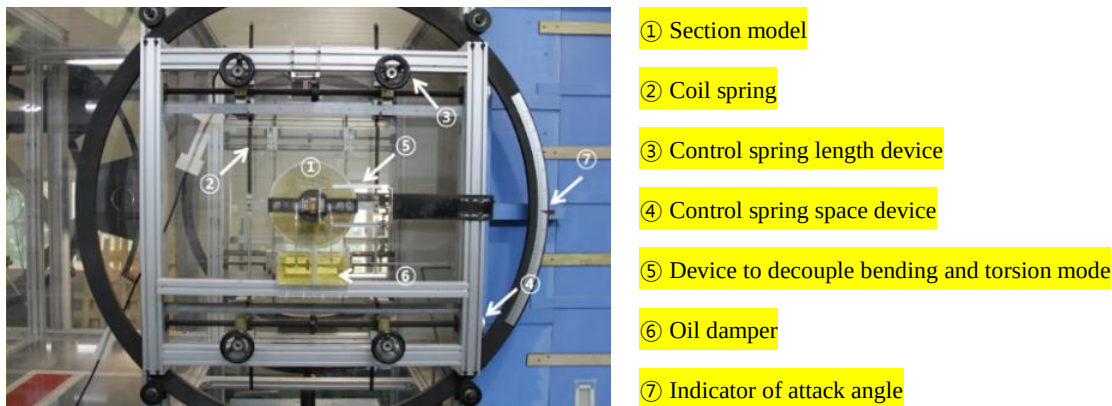


Figure 2. Spring support system of the wind tunnel

3.3 Data Acquisition Equipment

Displacements of the sectional model in the vibration tests were obtained by two laser displacement sensors. Two targets of laser sensor were attached at both sides of horizontal extension rods. The measured signals of model displacements

were filtered by low-pass filter, digitized by A/D converter, and stored in computer. The list of equipment used in this test is shown in Table 1.

Table 1. Data acquisition equipment list used in wind tunnel test

Measurements	Type of Sensor	Model Name (Company)	Capacity
Wind Velocity	Pitot Tube	LK-5 (Okano)	-
	Digital Manometer	FC012-14 (Furness)	19.99mmH ₂ O
Displacement	Laser Displacement Sensor	LB-1201/301(KEYENCE)	±10cm
Remove Noise Signal	Low Pass Filter	9B02 (NEC)	1Hz ~ 1kHz

3.4 Wind tunnel test procedure

The section model tests for aerodynamic response were conducted according to the procedure as follows. The wind speed is gradually increased in small steps from the lowest level available for the wind tunnel. The motions of section model in both the vertical and torsional directions are recorded. The wind speed is increased until the model attains flutter instability or beyond the design wind speed for the bridge. In order to clarify the phenomena such as vortex-induced vibrations or flutters, the test is conducted in smooth flow condition. Measurement results for the wind tunnel tests is presented with graphs of vertical and torsional mean, RMS and peak responses as a function of wind speed.

4. PARAMETER OPTIMIZATION FOR RINO

4.1 Overview

Figure 3 shows the measurement setup for the wind tunnel test. Two laser sensors and two iPhone 6 plus were installed for each to measure dynamic displacements and compare them. Each targets were installed at the L/2 position of horizontal extension rods supporting section model that enable sensors to measure actual displacement of section model and target shows target composed of two-patterns of RINO (see Figure 4). Each patterns had two-colors to be easily identified by RINO (Min et al, 2015), a sample target pattern is shown in Figure 3.

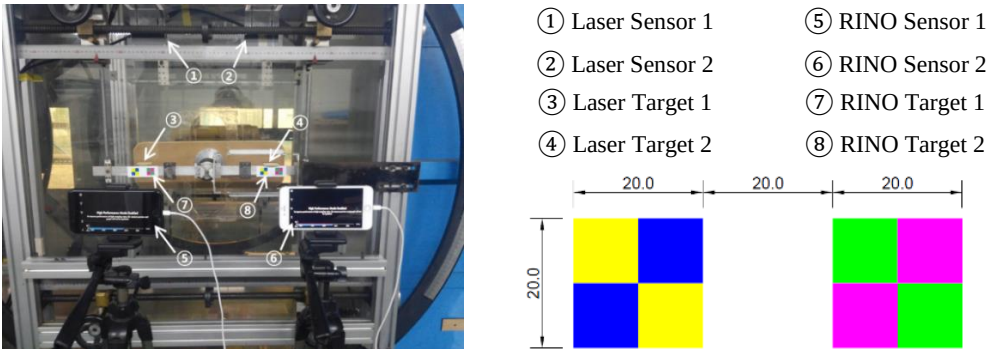


Figure 3. Setting of test model, equipment and target pattern

RINO offers sensing parameters that can be adjusted by users, including frame rate, crop filter size, and use of high performance mode. Table 2 shows adjustable setting details available for RINO. The vertical and horizontal size of pattern was 20mm and center-to-center spacing was 40mm. Measurements of *x* and *y* were displayed to screen and filter setting was default. Preliminary test was conducted to find optimized parameter for camera setting.

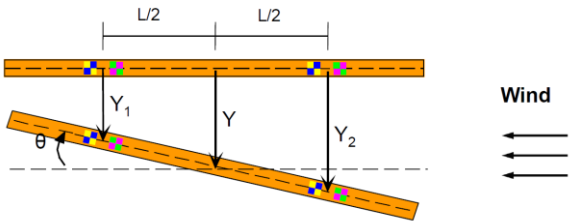


Figure 4. Definition of Displacement

4.2 Frame rate selection

Preliminary test has been conducted first to decide optimized frame rate under the 400 × 400 crop size and normal-performance condition by vertical mode and torsional mode of free vibration tests. Figure 8 shows time-dependent variation of vertical displacement measured by each two laser sensors and RINO. According to experimental results from laser and RINO agree well in cases of 30, 60, 120 and 240 fps conditions. However, recording data by 30 fps mode seems to have limitation for accurate measurement because of the natural frequencies of model that are 1.95Hz and 5.15Hz for bending and torsion. In case of 60 fps, all data points were clearly measured even without activating high performance mode.

This result showing that at least, 60 fps should be used for dynamic test. However, 120 and 240 fps case results shows no significant differences in measured data points to the 60 fps cases due to the missed data supposed to be incurred by speed limitation of GPU.

Table 2. Adjustable sensing parameters for RINO

Camera Settings		Graph Settings		Filter Settings	
Frame Rate	Adjust the camera’s frame rate 30, 60, 120 and 240 fps	Target Distance (cm)	The distance (in centimeters) of the two colored patterns in the camera frame	Pixel Search	The distance between neighboring pixels used in the OpenGL shader.
Crop Size (width & height)	Width (px): 100/200/400/720/1280 Height (px): 100/200/400/720	Show X & Y Displacement	Whether the X, Y, or both displacement lines should be shown on the graph.	Set Filter Colors	Select the hue ranges of the two colors used in each target.
High Performance	High Performance: on(without displaying), off(with displaying)				

4.3 High performance mode

Optimization test results shows under the high performance off condition, displaying screen occupies the GPU resources slowing down data calculation. Thus, 60 fps and 120 fps test data were compared in high-performance condition. In case of 60 fps, measurements by RINO agree well with laser sensor measurement and there were no missing data from high performance on/off condition. In case with 120 fps, data quantity from high performance condition was increased by nearly 50 percent than case of high performance off condition. Thus, over 120fps condition, high performance mode should be selected to acquire reasonable data.

4.4 Crop filter size

RINO offer crop filter to improve processing speeds. User can select crop area allowing minimizing processing area that drastically reduce processing time. Free vibration tests under 200 × 400 crop size which was minimal size which can hold displacement of two targets, 400 × 720 crop size which covers wide range were implemented and compared with the displacement of 400 × 400 crop size as shown in the Figure 5 to decide optimal crop size.

Displacements measured for each cases agree very well with those from the laser sensors. In case with 200 × 400 crop size and 400 × 400 crop size, there were no missing data but in case with 400 × 720 crop size, nearly half of data was recorded despite recorded in high performance mode. Thus, test condition is selected for 120 fps, 400 × 400 crop size condition considering processing speed and target movement area.

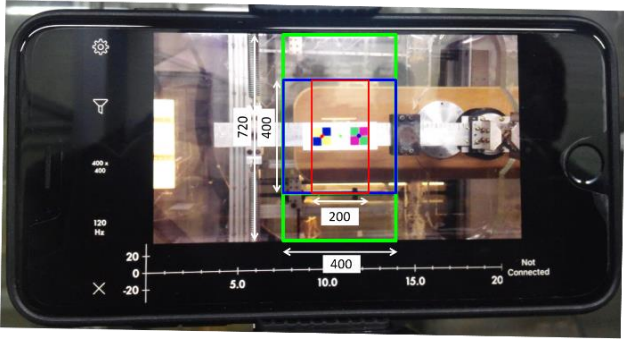


Figure 5 RINO screen and Crop size

5. WIND TUNNEL TEST RESULTS FOR A BRIDGE DECK: PERFORMANCE COMPARISON OF RINO VS. LASER SENSORS

5.1 Test condition

In this study, test bridge section as shown in Figure 6 was selected, where various wind-induced vibrations such as vortex shedding and flutter were observed. Section model test was implemented to compare the wind vibration measurement data from conventional laser sensor and RINO.

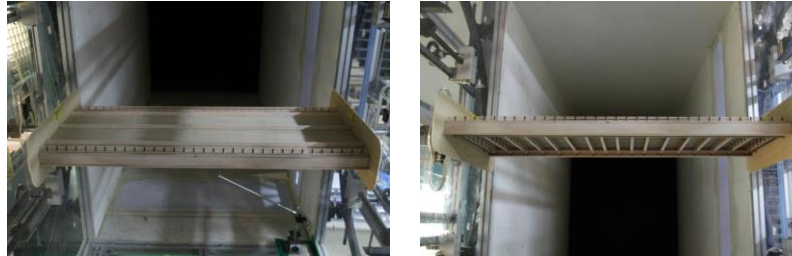


Figure 6. Section test model in the wind tunnel

The bridge deck section model mounted on the spring support system is shown in Figure 6 and the dynamic experimental characteristics of the model are shown in Table 3. Two measurement conditions were tested. First test was measured under the same condition with conventional setup using two RINO at both sides and the second test was measured by single RINO at one side to test performance of single RINO.

Table 3. Experimental Conditions

Item	Width (cm)	Weight (kg/m)	Polar Moment of Inertia(kg·cm ² /m)	Frequency (Hz)		Freq. ratio	Structural Damping(%)	
				Bending	Torsion		Bending	Torsion
Measured Value	34.4	4.89	599.4	1.95	5.15	2.64	0.28	0.13

5.2 Test results

Vertical and torsional displacement (Y , θ) were calculated by measured centroid displacement of two targets (Y_1 , Y_2) from RINO and laser sensor

$$Y = \frac{Y_1 + Y_2}{2} \quad (1)$$

$$\theta = \text{atan} \frac{Y_2 - Y_1}{L} \quad (2)$$

Figure 7 shows V-A Curve (Velocity-Amplitude Curve) from test result. The peak amplitude by vertical vortex-induced vibration is appear at 1.6m/s, torsional vortex-induced vibration is appearing at 2.2m/s and torsional flutter is appearing after 5m/s of wind velocity. Comparing with measurement of laser sensor, there was no significant difference with RINO at mean, RMS and Peak values. Figure 8 shows the time history at peak amplitude of vertical vortex-induced vibration, torsional vortex-induced vibration and torsional flutter. These two results from RINO and laser sensor showing equivalent performance to utilize in practical aerodynamic test measurement.

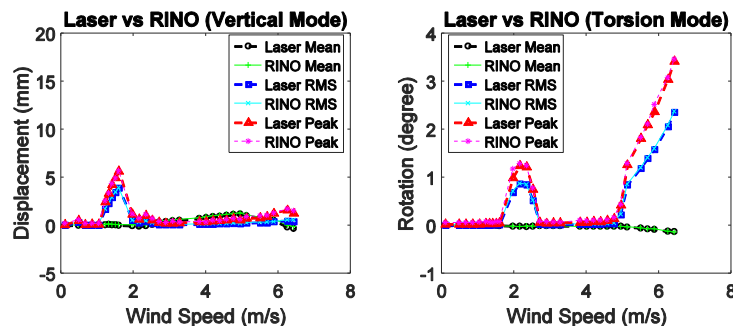
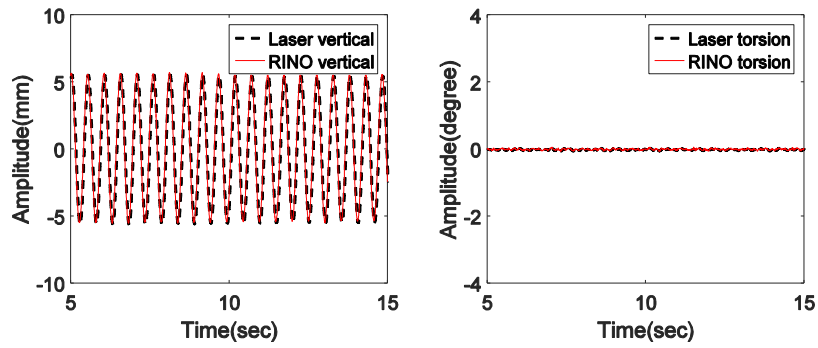
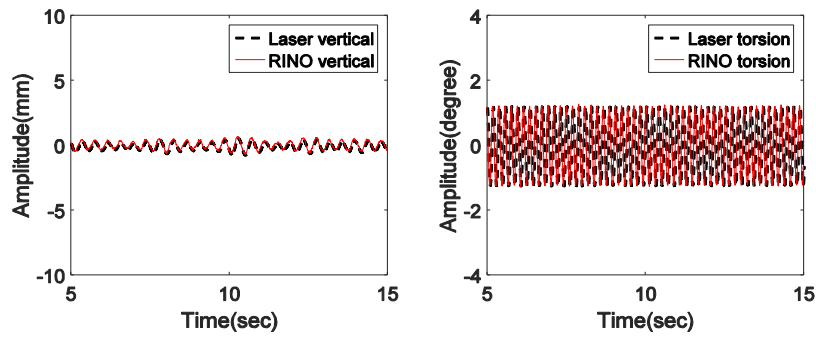


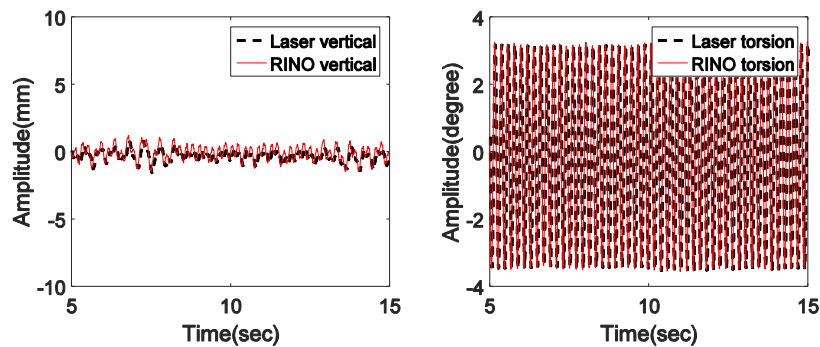
Figure 7. V-A Curve



(a) Maximum vertical vortex-induced vibration



(b) Maximum torsional vortex-induced vibration



(c) Maximum torsional flutter

Figure 8. Time history response

RINO offer vertical and horizontal displacement from centroid of target as well as each patterns. Vertical and torsional displacement of test model was calculated by using centroid measurements of two-patterns of one-pattern as follows (see Figure 9).

$$Y = \frac{y_2(L_1 - L_2) - y_1L_2}{L_1} \quad (3)$$

$$\theta = \text{asin}\left(\frac{y_2 - y_1}{L_1}\right) \quad (4)$$

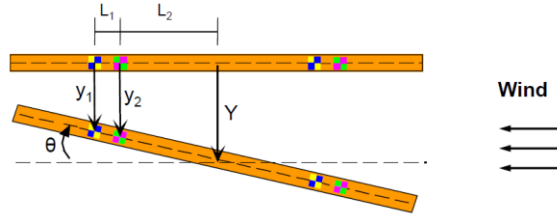


Figure 9. Torsional displacement estimation using RINO

Measurements from two-laser sensor and a RINO sensor are shown in Figure 10 that mean and RMS values agree well. However, the peak value from a RINO sensor was generally larger than the values from laser sensor. Figure 11 shows the time history at peak amplitude of vertical vortex-induced vibration, torsional vortex-induced vibration and torsional flutter. Despite noises at +/- peak of each cycle were observed in the measurement of RINO, overall results agree well.

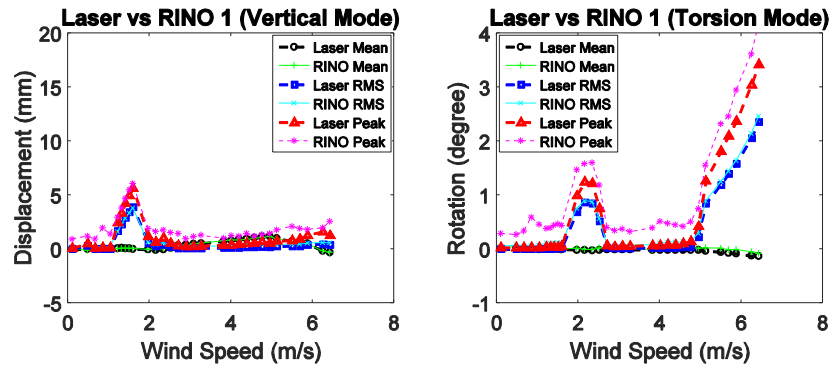
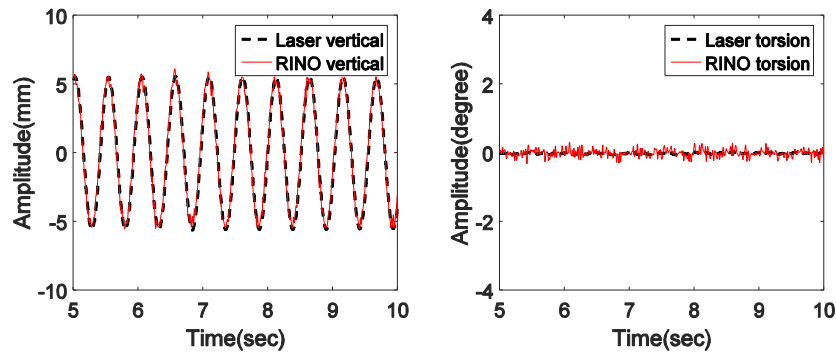
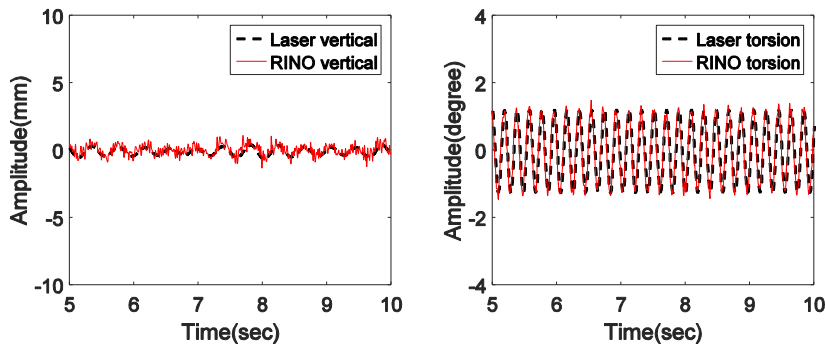


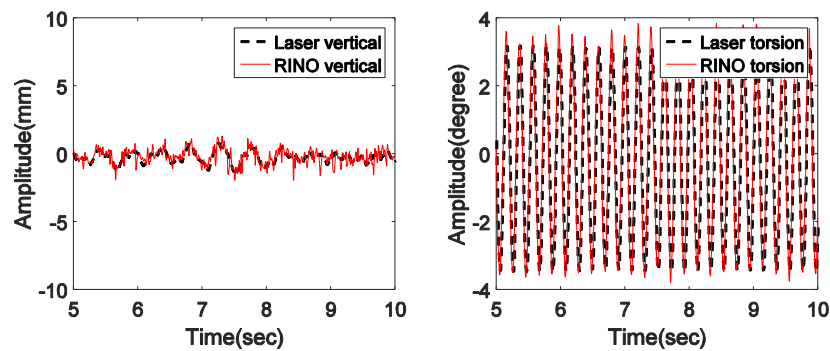
Figure 10. Comparison of V-A Curves



(a) Maximum vertical vortex-induced vibration



(b) Maximum torsional vortex-induced vibration



(c) Maximum torsional flutter

Figure 11 Time history response

5.3 Discussion

Dynamic displacements of a bridge deck section model during wind-tunnel tests have been measured by two RINOs and a single RINO and compared with those from two-laser sensors to see the feasibility of RINO for wind-tunnel test. Measurements from two RINOs offer equivalent level of accuracy to laser sensors for mean, RMS and peak values. Measurements by a RINO show some noises from target tracking incurred difference peak values while mean and RMS values shows no significant difference values from laser sensors. Tests show that estimating displacements from two pattern significantly reduces tracking noises, thus act as a mean filter. In this study, any other filters were not adopted to reduce noise thus, measurements from a single RINO are expected to be further improved by implementing various noise filters. As a result, RINO provides practically equivalent method to measure dynamic motion of model structure in aerodynamic tests. Although conventional laser displacement sensor shows better quality in many aspects such as resolution and sampling rate, RINO has significant advantages in various aspects in win tunnel test, including cost efficiency, accuracy and usability that a RINO can replace four-lasor sensor by measuring each displacement of two patterns.

6. CONCLUSION

The feasibility of iOS application RINO for wind-tunnel test has been investigated. Test measurements from RINO were compared with laser sensor. In order to investigate feasibility of utilizing RINO for practical test measurement, wind-tunnel tests were conducted under the same measurement condition that each two lessor sensors and RINO measured same position of external rods connected with bridge section model. Before the practical tests, several preliminary tests were conducted to find optimized user parameter of RINO for wind-tunnel test. Performances by frame rate, high performance mode and crop filter sizes were investigated comparing with commercially used laser sensors to decide best setting to measure aerodynamic displacement from section model. From preliminary test, high performance mode, 400x400 crop filter and 120 fps were selected as user parameter for wind-tunnel test. Measurements from two RINOs shows no significant difference with commercial laser sensor. Additional wind-tunnel test was conducted to investigate the measurement from a single RINO. The results shown that measurement from a single RINO were comparable to laser sensor except peak values induced by tracking noise. The possibilities and limitations of RINO for aerodynamic wind-tunnel tests were investigated. Test results from RINO have shown comparable measurements to those of commercial laser sensor, but also some limitations were observed using a single RINO, which are supposed to be further improved by following research.

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